SYSTEM CALLS

```
Activities  Terminal                Mar 31  11:34
suhrithasri@localhost:~/system_calls
[suhrithasri@localhost:~]$ ls
abc.sh   Desktop   Downloads   ifelse.sh  Music   Pictures   Public   REC      sample.sh   scheduling  sh     Templates  while.sh
ab.sh    Documents Folder      loops.sh   pc      posneg.sh  qns.txt  sample.c sample.txt  SFA       system_calls  Videos
[suhrithasri@localhost ~]$ cd system_calls
[suhrithasri@localhost system_calls]$ vi pid.c
[suhrithasri@localhost system_calls]$ gcc pid.c
[suhrithasri@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[suhrithasri@localhost system_calls]$ vi fork.c
```

```
Activities  Terminal                Mar 31  11:35
suhrithasri@localhost:~/system_calls — /usr/bin/vim fork.c
#include <stdio.h>
#include <unistd.h>

int main(){
        if(fork()==0){
                printf("This is CHILD process\n");
        } else {
                printf("This is PARENT process\n");
        }

        printf("PID: %d\n", getpid());

        return 0;
}
~
~
~
~
~
```

```
Activities  Terminal                Mar 31  11:37
suhrithasri@localhost:~/system_calls
suhrithasri@localhost:~$ ls
abc.sh   Desktop   Downloads   ifelse.sh  Music   Pictures   Public   REC      sample.sh   scheduling  sh     Templates  while.sh
ab.sh    Documents Folder      loops.sh   pc      posneg.sh  qns.txt  sample.c sample.txt  SFA       system_calls  Videos
[suhrithasri@localhost ~]$ cd system_calls
[suhrithasri@localhost system_calls]$ vi pid.c
[suhrithasri@localhost system_calls]$ gcc pid.c
[suhrithasri@localhost system_calls]$ ./a.out
PID: 2802
PPID: 2505
[suhrithasri@localhost system_calls]$ vi fork.c
[suhrithasri@localhost system_calls]$ gcc fork.c
[suhrithasri@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[suhrithasri@localhost system_calls]$ vi exec.c
```

```
Activities  Terminal                    Mar 31  11:31
                    suhrithasri@localhost:~/system_calls
[suhrithasri@localhost ~]$ ls
abc.sh   Desktop    Downloads   ifelse.sh  Music   Pictures   Public   REC       sample.sh   scheduling  sh        Templates  while.sh
ab.sh    Documents  Folder      loops.sh   pc      posneg.sh  qns.txt  sample.c  sample.txt  SFA         system_calls Videos
[suhrithasri@localhost ~]$ cd system_calls
[suhrithasri@localhost system_calls]$ vi pid.c
```

```
Activities  Terminal                    Mar 31  11:39
          suhrithasri@localhost:~/system_calls — /usr/bin/vim exec.c
#include <stdio.h>
#include <unistd.h>

int main(){
        printf("Before exec()\n");
        execl("/bin/ls", "ls", NULL);
        printf("After exec()\n");

        return 0;
}
~
~
~
~
```

```
Activities  Terminal                    Mar 31  11:41
                    suhrithasri@localhost:~/system_calls
suhriithasri@localhost:~l$ ls
abc.sh   Desktop    Downloads   ifelse.sh  Music   Pictures   Public   REC       sample.sh   scheduling  sh        Templates  while.sh
ab.sh    Documents  Folder      loops.sh   pc      posneg.sh  qns.txt  sample.c  sample.txt  SFA         system_calls Videos
suhrithasri@localhost:~$ cd system_calls
suhrithasri@localhost:~/system_calls$ vi pid.c
suhrithasri@localhost:~/system_calls$ gcc pid.c
suhrithasri@localhost:~/system_calls$ ./a.out
PID: 2802
PPID: 2505
suhrithasri@localhost:~/system_calls$ vi fork.c
suhrithasri@localhost:~/system_calls$ gcc fork.c
suhrithasri@localhost:~/system_calls$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
suhrithasri@localhost:~/system_calls$ vi exec.c
suhrithasri@localhost:~/system_calls$ gcc  exec.c
suhrithasri@localhost:~/system_calls$ ./a.out
Before exec()
a.out  exec.c  fork.c  opendir.c  pid.c  readdir.c
suhrithasri@localhost:~/system_calls$ vi opendir.c.:c
```

```
Activities  Terminal                    Mar 31  11:41
          suhrithasri@local host: ~/system_calls — /usr/bin/vim opendir.c
#include <stdio.h>
#include <dirent.h>

int main(){
        DIR *dir;

        dir = opendir(".");

        if(dir == NULL){
                printf("Failed to open directory\n");
                return 1;
        }

        printf("Directory opened successfully!\n");
        closedir(dir);
        return 0;
}
~
~
~
```

Activities Terminal Mar 31 11:42

suhrithasri@local host: ~/system_calls

```
suhrithasri@local host:~$ ls
abc.sh  Desktop   Downloads  ifelse.sh  Music  Pictures  Public  REC      sample.sh  scheduling  sh          Templates  while.sh
ab.sh   Documents Folder     loops.sh   PC     posneg.sh  qns.txt sample.c sample.txt SFA         system_calls Videos
suhrithasri@local host:~$ cd system_calls
suhrithasri@local host:~/system_calls$ vi pid.c
suhrithasri@local host:~/system_calls$ gcc pid.c
suhrithasri@local host:~/system_calls$ ./a.out
PID: 2802
PPID: 2505
suhrithasri@local host:~/system_calls$ vi fork.c
suhrithasri@local host:~/system_calls$ gcc fork.c
suhrithasri@local host:~/system_calls$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
suhrithasri@local host:~/system_calls$ vi exec.c
suhrithasri@local host:~/system_calls$ gcc exec.c
suhrithasri@local host:~/system_calls$ ./a.out
Before exec()
a.out  exec.c  fork.c  opendir.c  pid.c  readdir.c
suhrithasri@local host:~/system_calls$ vi opendir.c
suhrithasri@local host:~/system_calls$  gcc opendir.c
suhrithasri@local host:~/system_calls$ ./a.out
Directory opened successfully!
suhrithasri@local host:~/system_calls$ vi  readdir.c
```

Activities Terminal Mar 31 11:44

suhrithasri@local host:~/system_calls — /usr/bin/vim readdir.c

```c
#includde <stdio.h>
#includde <dirent.h>

int main(){
        DIR *dir;
        struct dirent *entry;

        dir = opendir(".");

        if(dir == NULL){
                printf(Cannot open directory\n");
                return 1;
        }

        printf("Files in directory:\n");
        while((entry*=readdir(dir)) != NULL){
                printf("%s\n", entry->d_name);
        }

        closedir(dir);

        return 0;
}
~
~
```

Activities Terminal Mar 31 11:45

suhrithasri@local host:~/system_calls

```
suhrithasri@local host:~$ ls
abc.sh  Desktop   Downloads  ifelse.sh  Music  Pictures  Public  REC      sample.sh  scheduling  sh          Templates  while.sh
ab.sh   Documents Folder     loops.sh   PC     posneg.sh  qns.txt sample.c sample.txt SFA         system_calls Videos
suhrithasri@local host:~$ cd system_calls
suhrithasri@local host:~/system_calls$ vi pid.c
suhrithasri@local host:~/system_calls$ gcc pid.c
suhrithasri@local host:~/system_calls$ ./a.out
PID: 2802
PPID: 2505
suhrithasri@local host:~/system_calls$ vi fork.c
suhrithasri@local host:~/system_calls$ gcc fork.c
suhrithasri@local host:~/system_calls$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
suhrithasri@local host:~/system_calls$ vi exec.c
suhrithasri@local host:~/system_calls$ gcc exec.c
suhrithasri@local host:~/system_calls$ ./a.out
Before exec()
a.out  exec.c  fork.c opendir.c  pid.c readdir.c
suhrithasri@local host:~/system_calls$ vi opendir.c
suhrithasri@local host:~/system_calls$  gcc opendir.c
suhrithasri@local host:~/system_calls$  ./a.out
Directory opened successfully!
suhrithasri@local host:~/system_calls$ vi readdir.c
suhrithasri@local host:~/system_calls$  gcc readdir.c
suhrithasri@local host:~/system_calls$ ./a.out
Files in directory:
pid.c
fork.c
exec.c
opendir.c
readdir.c
a.out
suhrithasri@local host:~/system_calls$
```